Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 PHYSICS

9478/01

22 September 2026

1 hour

Paper 1 Multiple Choice

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **thirty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct
- and record your choice in **soft pencil** on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and index number on the multiple choice answer sheet in the
- spaces provided unless this has been done for you.
- Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 30.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

Data

speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\,\text{H m}^{-1}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$
	$(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31}\,\text{kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27}\,\text{kg}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Avogadro constant	$N_A = 6.02 \times 10^{23}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T/\text{K} = T/\,^\circ\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

© DHS 2026

displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + ...$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + ...$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + ...$
capacitors in parallel	$C = C_1 + C_2 + ...$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Suggested Ans

1. C	2. A	3. B	4. D	5. C
6. A	7. B	8. D	9. D	10. A
11. C	12. B	13. B	14. A	15. A
16. D	17. B	18. B	19. C	20. B
21. B	22. A	23. A	24. C	25. C
26. D	27. B	28. C	29. B	30. B

1 After taking measurements of the quantities in the expression $\frac{3PQ^4}{R^2}$, the total uncertainty is calculated as 11%.

Which individual percentage uncertainties of P, Q and R when combined give this total of 11%?

	***P* / %**	***Q* / %**	***R* / %**	z / %
A	1.0	1.0	2.0	**9.0**
B	2.0	2.0	1.5	**13**
C	3.0	1.0	2.0	**11**
D	5.0	2.0	1.0	**15**

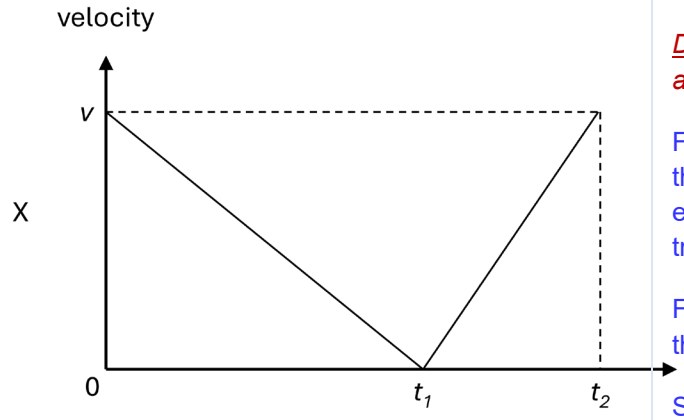
Ans: C

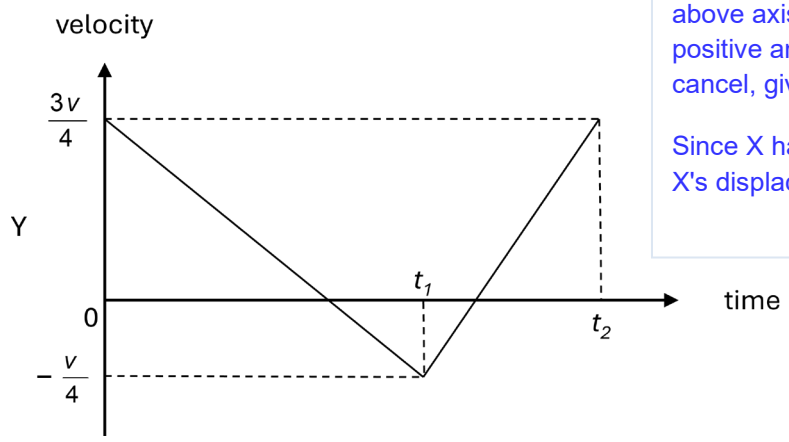
Let $z = \frac{3PQ^4}{R^2}$

$\frac{\Delta z}{z} = \frac{\Delta P}{P} + 4\frac{\Delta Q}{Q} + 2\frac{\Delta R}{R}$

C is the only option which gives 11%

2 The motions of two objects X and Y along a straight line are shown in the following velocity-time graphs.





On a velocity-time graph:

Displacement = area above the time axis minus area below the time axis

For X, the entire graph sits above the time axis, so the displacement equals the total area of the two triangles above the axis.

For Y, some of the graph is below the time axis.

So, the net displacement = (area above axis) − (area below axis). The positive and negative areas partially cancel, giving a smaller net result.

Since X has no cancellation at all, X's displacement is larger.

How do the magnitudes of the displacement and the total distance travelled compare for the two objects from time 0 to time t_2?

Ans: A

	displacement	**distance**

6

A	different	different
B	different	same
C	same	different
D	same	same

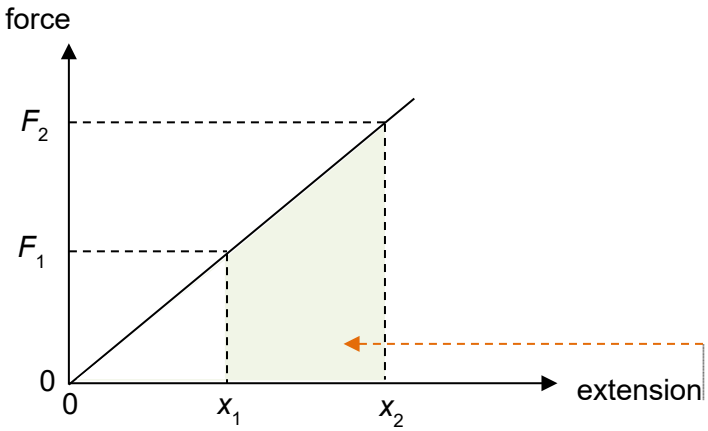
On a velocity-time graph:

Total distance = area above + |area below| (all areas treated as positive)

For X, since the graph never goes negative, total distance = displacement = same total area as before.

For Y, you'd add the area below the axis back in (instead of subtracting it). However, the key insight is that Y's graph reaches a lower peak on the positive side (only 3*v*/4 compared to X's *v*) and the negative dip "wastes" some of the area on cancellation. After adding up, the total area for Y still comes out smaller than X's.

3 The force-extension graph of a spring is as shown.



What is the change in elastic potential energy when the extension of the spring is increased from x_1 to x_2?

A $F_2x_2 - F_1x_1$

B $\frac{1}{2}(F_2x_2 - F_1x_1)$

C $F_2x_2^2 - F_1x_1^2$

D $\frac{1}{2}(F_2x_2^2 - F_1x_1^2)$

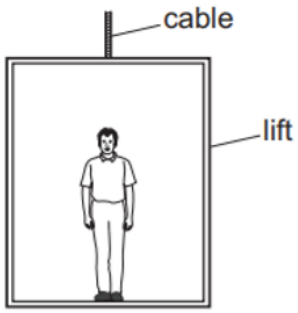
Ans: B

The area under the force-extension graph gives the elastic potential energy stored in the spring.

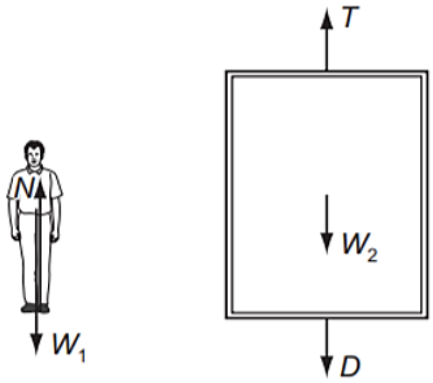
As extension is increased from x_1 to x_2, the area changes from $\frac{1}{2}F_1x_1$ to $\frac{1}{2}F_2x_2$.

Hence the change in EPE = $\frac{1}{2}(F_2x_2 - F_1x_1)$.

4 The diagram shows a man standing in a lift.



The forces acting on the man and the forces acting on the lift are shown below.



N is the force from the lift floor on the man.
W_1 is the weight of the man.
T is the tension in the lift cable.
W_2 is the weight of the lift.
D is the force from the man on the lift floor.

Which of the following statements is correct?

A $(W_1 + W_2)$ is always equal to T.

B If $N = W_1$, the lift must be at rest.

C N and W_1 are always equal and opposite.

D If $T = (D + W_2)$, the lift must have a constant velocity.

A: This would only be true if the lift system has zero acceleration. If the lift accelerates, T must be greater or less than $(W_1 + W_2)$. So, this is incorrect.

B: For the man: $N - W_1 = ma$. If $N = W_1$, then $a = 0$, meaning zero acceleration — but this includes constant velocity motion, not just rest. So, this is incorrect.

C: N and W_1 are only equal when acceleration is zero. When the lift accelerates, $N \neq W_1$. Also, they are not a Newton's 3rd law pair (they both act on the same object — the man). So, this is incorrect.

Ans: D

Applying Newton's 2nd law to the **lift alone**:

$$T - W_2 - D = m_{\text{lift}} \times a$$

If $T = D + W_2$, then the net force on the lift = 0, meaning $\boldsymbol{a = 0}$, so the lift moves at constant velocity (or is at rest). This is **correct**.

5 A trolley runs down a slope with a constant acceleration *a*. The mass of the trolley is now doubled, and the trolley is allowed to run down the same slope. In both cases effects of friction and air resistance are negligible.

A The accelerating force is the same.

B The acceleration is $\frac{1}{2}a$.

C The acceleration is *a*.

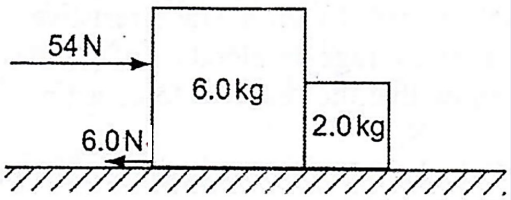
D The acceleration is 2*a*.

Ans: C
For a trolley on a frictionless slope, the only force causing acceleration is the component of gravity along the slope: $F = mg\sin\theta$
The acceleration is: $a = \frac{F}{m} = \frac{mg\sin\theta}{m} = g\sin\theta$
When Mass is Doubled
The new accelerating force becomes: $F' = 2mg\sin\theta$
So, the accelerating force is doubled — meaning option A is incorrect.
The new acceleration is:

$$a' = \frac{F'}{2m} = \frac{2mg\sin\theta}{2m} = g\sin\theta = a$$

The mass cancels out completely, just like free fall.

6 A force of 54 N pushes two touching blocks of mass 6.0 kg and 2.0 kg along a flat surface. The frictional force between the blocks and the surface is 6.0 N.



What is the force exerted on the 2.0 kg block?

A 12 N **B** 36 N **C** 45 N **D** 48 N

Ans: A
Applying Newton's 2nd law to the entire system: $F_{net} = 54 - 6.0 = 48$ N

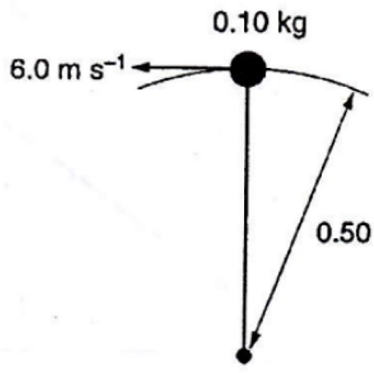
$$a = \frac{F_{net}}{m_{total}} = \frac{48}{8.0} = 6.0 \text{ m s}^{-2}$$

Now consider the 2.0 kg block alone. The only horizontal force acting on it is the contact force from the 6.0 kg block pushing it forward (there is no separate friction stated for the 2.0 kg block individually — the 6.0 N friction acts on the whole system).

Applying Newton's 2nd law to the 2.0 kg block:

$$F = ma = 2.0 \times 6.0 = 12 \text{ N}$$

7 A ball of mass 0.10 kg is attached to a string and swung in a vertical circle of radius 0.50 m, as shown. Its speed at the top of the circle is 6.0 m s^{-1}.



What is the tension in the string at this moment?

A 0.98 N **B** 6.2 N **C** 7.2 N **D** 8.2 N

Ans: B
At the **top** of the vertical circle, both the tension *T* and the weight *W* ac
downward toward the centre of the circle, providing the centripetal for

$$T + W = \frac{mv^2}{r}$$

Substituting values
centripetal force:

$$\frac{mv^2}{r} = \frac{0.10 \times 6.0^2}{0.50} = \frac{0.10 \times 36}{0.50} = 7.2 \text{ N}$$

weight:

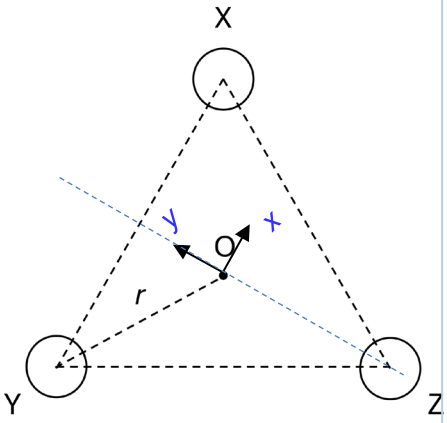
$$W = mg = 0.10 \times 9.81 = 0.98 \text{ N}$$

solving for tension:

$$T = \frac{mv^2}{r} - W = 7.2 - 0.98 = 6.22 \text{ N} \approx 6.2 \text{ N}$$

Use the following diagram to answer questions 8 and 9.

Three planets, X, Y and Z have masses *M*, *M* and *2M* respectively, are arranged in a triangle and are equidistant from each other.



Q8. Ans: D

Each planet contributes a gravitational field at O of magnitude:

$$g = \frac{GM}{r^2}, \quad g_Y = \frac{GM}{r^2}, \quad g_Z = \frac{G(2M)}{r^2} = \frac{2GM}{r^2}$$

Since the triangle is equilateral, the fields from X and Y point toward X and Y respectively. By symmetry, the **horizontal components** of X and Y's fields cancel each other out, and their **vertical components** (pointing upward, away from Z) add together.

The vertical component of each (X and Y) pointing away from Z:

$$g_X \cos 30° + g_Y \cos 30° = 2 \times \frac{GM}{r^2} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}GM}{r^2}$$

The field from Z points **toward Z** (downward):

$$g_Z = \frac{2GM}{r^2}$$

Net field:

$$g_{net} = \frac{2GM}{r^2} - \frac{\sqrt{3}GM}{r^2} = \frac{GM(2-\sqrt{3})}{r^2} = \frac{G(2M-\sqrt{3}M)}{r^2}$$

Since 2 > √3, the net field points **toward Z**.

8 What is the magnitude and direction of the gravitational field strength at point O?

	magnitude	**direction**
A	0	no direction
B	$\frac{GM}{r^2}$	towards X
C	$\frac{GM}{r^2}$	towards Z
D	$\frac{G\left(2M-\sqrt{3}M\right)}{r^2}$	towards Z

9 What is the gravitational potential at O and its direction?

	potential	**direction**
A	0	no direction
B	$-\frac{GM}{r}$	towards Z
C	$\frac{GM}{r}$	towards Z
D	$-\frac{4GM}{r}$	no direction

Q9. Ans: D

Gravitational potential is a **scalar** quantity, so we simply add the contributions from all three planets without worrying about directions.

$$\phi = -\frac{GM}{r} - \frac{GM}{r} - \frac{G(2M)}{r}$$

$$\phi = -\frac{4GM}{r}$$

Gravitational potential is always **negative** and is a scalar — it has **no direction**. Looking at the options, option D gives magnitude $\frac{4GM}{r}$ with no direction, which matches (the negative sign is implied by the nature of gravitational potential).

10 A small mass on a vibrating platform undergoes simple harmonic motion. The total energy of the mass is 18 mJ.

The amplitude of the oscillation is now doubled, and the time period is increased to be three times as long.

What is the new total energy of the mass?

A 8 mJ **B** 12 mJ **C** 36 mJ **D** 72 mJ

Ans: A

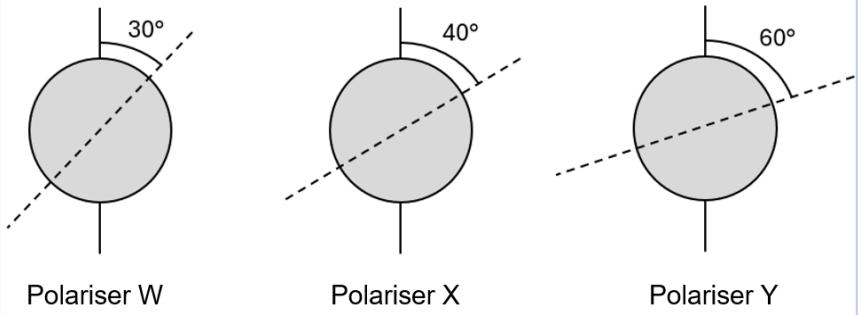
total energy of a simple harmonic oscillator,

$$E_{total} = \frac{1}{2}mv_0^2 = \frac{1}{2}m(\omega x_0)^2 \propto \left(\frac{x_0}{T}\right)^2 \Rightarrow \frac{E_{new}}{E_{old}} = \left(\frac{2}{3}\right)^2 = \frac{4}{9} \Rightarrow E_{new} = \frac{4}{9}E_{old} = \frac{4}{9}(18\text{ mJ}) = 8\text{ mJ}$$

11 The diagram below shows three polarisers W, X and Y.

A beam of unpolarised light of intensity I_0 is incident normally onto the surface of polariser W, which then passes through polariser X and eventually emerges from polariser Y.

The axis of polarisation of each polariser is indicated by a dashed line.



Ans: C
Angle between successive polarisers:
- W to X: 40° − 30° = 10°
- X to Y: 60° − 40° = 20°

After Polariser W
Unpolarised light passing through the first polariser loses half its intensity: $I_W = \frac{I_0}{2} = 0.5I_0$

After Polariser X (Malus's Law)

$I_X = I_W \cos^2(10°)$
$= 0.5I_0 \times \cos^2(10°)$
$= 0.5I_0 \times 0.9698$
$= 0.4849I_0$

After Polariser Y (Malus's Law)

$I_Y = I_X \cos^2(20°)$
$= 0.484I_0 \times \cos^2(20°)$
$= 0.484I_0 \times 0.8830$
$= 0.428I_0 \approx 0.43I_0$

What is the intensity of the light that emerges from polariser Y?

A $0.11\,I_0$ **B** $0.32\,I_0$ **C** $0.43\,I_0$ **D** $0.64\,I_0$

12 A light wave of amplitude *A* is incident normally on a surface of area *S*. The intensity reaching the surface is I.

The amplitude of the light wave is increased to 2*A*. The light is then focused on to a smaller area $\frac{1}{3}S$.

What is the intensity on this smaller area?

A $6I$ **B** $12I$ **C** $18I$ **D** $24I$

Ans: B
Intensity (power per unit area) is proportional to the square of amplitude: $I \propto A^2$

Effect of doubling the amplitude
Original intensity (power per unit area) = I, with amplitude *A*.
When amplitude is doubled to 2*A*: $I_{new} = k(2A)^2 = 4kA^2 = 4I$

Effect of focusing onto smaller area
The same total power is now focused onto area ⅓*S*:
New power per unit area $= {}^{4I \times S}/_{\frac{1}{3}S} = 4I \times 3 = 12I$

© DHS 2026

13 The diameter of the receiving dish of a telescope is 5.1 m. It receives light of wavelength 0.40 μm from two point sources at a distance of 2.8×10^{25} m away.

What is the minimum separation of the two points sources so that the light from them can be resolved?

A 1.1×10^{18} m

B 2.2×10^{18} m

C 1.8×10^{32} m

D 3.6×10^{32} m

Ans: B

The minimum angular separation for two point sources to be resolved is: $\theta_{min} \approx \frac{\lambda}{b}$ where b is the diameter of the receiving dish.

minimum angle:

$$\theta_{min} \approx \frac{0.40\times10^{-6}}{5.1} = 7.843\times10^{-8} \text{ rad}$$

minimum separation:

For small angles, the separation x at distance D is: $x = \theta_{min} \times D$

$$x = 7.843\times10^{-8}\times2.8\times10^{25} \approx 2.2\times10^{18} \text{ m}$$

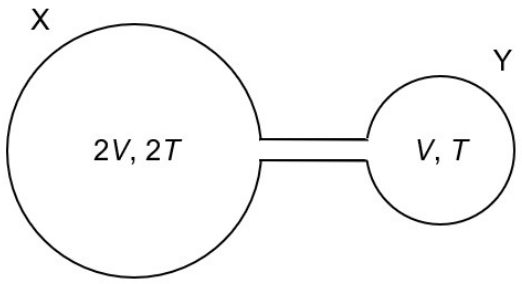
14 An ideal gas has thermodynamic temperature T, and its molecules has root-mean-square speed c_{rms}. If the gas is heated to temperature $2T$, what is the new root-mean-square speed of the molecules?

A $\sqrt{2}c_{rms}$ **B** $2c_{rms}$ **C** $2\sqrt{2}c_{rms}$ **D** $4c_{rms}$

Ans: A

$$\frac{1}{2}mc_{rms}^2 = \frac{3}{2}kT \Rightarrow c_{rms}^2 \propto T \Rightarrow \frac{T}{2T} = \frac{c_{rms}^2}{c_{rms,\,new}^2} \Rightarrow c_{rms,\,new} = \sqrt{2}c_{rms}$$

15 An ideal gas is contained in two spherical containers X and Y of volume $2V$ and V respectively, connected by a hollow tube of negligible volume. The containers X and Y are maintained at temperatures $2T$ and T respectively. The setup is shown in the figure below.



Determine the ratio

$$\frac{\text{number of moles of gas in container X}}{\text{number of moles of gas in container Y}}.$$

A 0.25 **B** 1.00 **C** 2.00 **D** 4.00

Ans: A

Since the containers are connected, the gas reaches the same pressure P throughout the system at equilibrium.

Using the ideal gas law: $pV = nRT$, so: $n = \frac{pV}{RT}$

Moles in Container X: $n_X = \frac{p\times2V}{R\times2T} = \frac{2pV}{2RT} = \frac{pV}{RT}$; Moles in Container Y: $n_Y = \frac{p\times V}{R\times T} = \frac{pV}{RT}$

Ratio: $\frac{n_X}{n_Y} = \frac{\frac{pV}{RT}}{\frac{pV}{RT}} = 1.00$

16 A 100 g block of ice at −10.0 °C is placed into 150 g of water initially at 20.0 °C inside a perfectly insulated calorimeter. What is the final temperature of the mixture and the mass of ice remaining?

Specific heat capacity of water = 4180 J kg^{-1} K^{-1}

Specific latent heat of fusion of ice = 334000 J kg^{-1}

Specific heat capacity of ice = 2100 J kg^{-1} K^{-1}

A 0 K with 31.3 g of ice remaining

B 0 K with 68.7 g of ice remaining

C 0 °C with 31.3 g of ice remaining

D 0 °C with 68.7 g of ice remaining

Ans: D

Heat available from water cooling to 0°C:

$$Q_{water} = mc\Delta T = 0.150 \times 4180 \times 20.0 = 12540 \text{ J}$$

Heat needed to warm ice from −10.0°C to 0°C:

$$Q_{ice} = mc\Delta T = 0.100 \times 2100 \times 10.0 = 2100 \text{ J}$$

Remaining heat available to melt ice:

$$Q_{remaining} = 12540 - 2100 = 10440 \text{ J}$$

Mass of ice melted:

$$m = \frac{10440}{334000} = 0.03126 \text{ kg} = 31.3 \text{ g}$$

Ice remaining:

$$100 - 31.3 = 68.7 \text{ g}$$

17 An electric current of 12 A is in a wire of length 0.35 m.

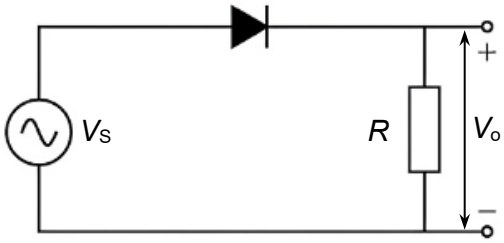
The average drift speed of the free electrons (charge carriers) in the wire is 5.0×10^{-4} m s^{-1}.

How many free electrons are in the wire?

A 1.3 × 10^{16} **B** 5.3 × 10^{22} **C** 1.5 × 10^{23} **D** 4.3 × 10^{23}

Ans: B $I = nAvq \Rightarrow \left(\frac{N}{V}\right)Avq = \frac{N}{L}vq \Rightarrow N = \frac{IL}{vq} = \frac{(12)(0.35)}{(5.0\times10^{-4})(1.60\times10^{-19})} = 5.3\times10^{22}$

18 An ideal semiconductor diode is connected in series with a load resistor R = 50 Ω and an alternating voltage source $V_S = 12.0\sin(100\pi t)$. The output voltage V_o is measured across the load resistor R.



Ans: B

The diode only conducts during the positive half-cycle, so current flows through R for only half the cycle.

Peak voltage across R:

$$V_0 = 12.0 \text{ V}$$

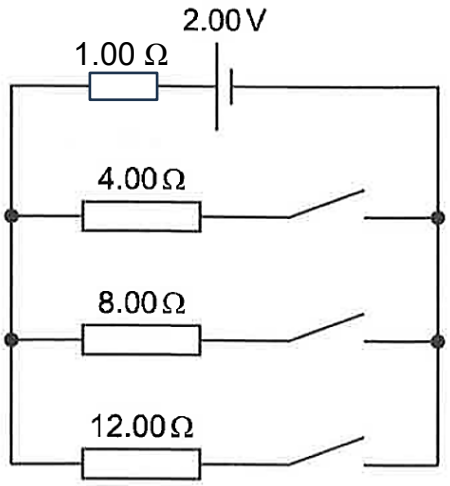
Average power over a full cycle: Since the diode conducts only for half the cycle, the average power is half that of a normal AC circuit:

$$\langle P \rangle = \frac{1}{2}\left(\frac{V_0^2}{2R}\right) = \frac{V_0^2}{4R} = \frac{12.0^2}{4 \times 50} = \frac{144}{200} = 0.72 \text{ W}$$

What is the average power dissipated in the load resistor R over one complete cycle?

A 0 W **B** 0.72 W **C** 1.44 W **D** 2.88 W

19 A cell of e.m.f. 2.00 V and internal resistance 1.00 Ω is connected to three resistors of 4.00 Ω, 8.00 Ω and 12.00 Ω as shown in the circuit diagram.



Ans: C

When every switch is closed, all three resistors are connected in parallel across the 2.00 V cell.

$$\frac{1}{R_{parallel}} = \frac{1}{4} + \frac{1}{8} + \frac{1}{12} = \frac{11}{24}$$

$$R_{parallel} = \frac{24}{11} = 2.182 \text{ Ω}$$

The total resistance R_{total} becomes 24/11 + 1.00 = 35/11 Ω

$$I = \frac{V}{R_{total}} = \frac{2.00}{\frac{35}{11}} = \frac{2.00 \times 11}{35} = 0.629 \text{ A}$$

What is the current through the cell when every switch is closed?

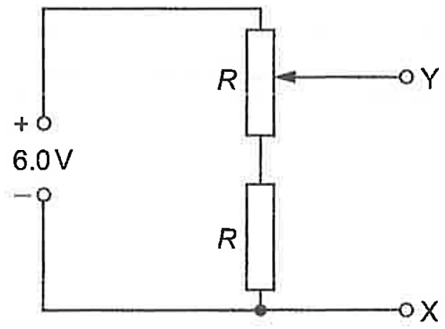
A 0.0800 A **B** 0.500 A **C** 0.629 A **D** 0.917 A

20 A potential divider has a constant supply of 6.0 V, as shown in the diagrams.

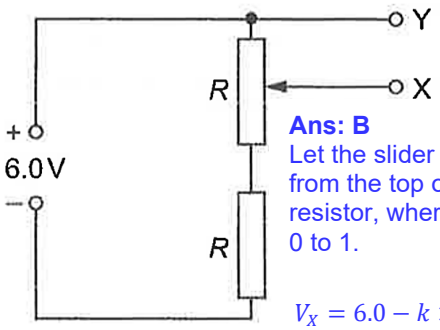
Which circuit will provide a potential difference between X and Y that can be varied between zero and 3.0V?

A

V_{XY} range $= \mathbf{3.0\ V\ to\ 6.0\ V}$



B



Ans: B

Let the slider be at a fraction k from the top of the upper resistor, where k ranges from 0 to 1.

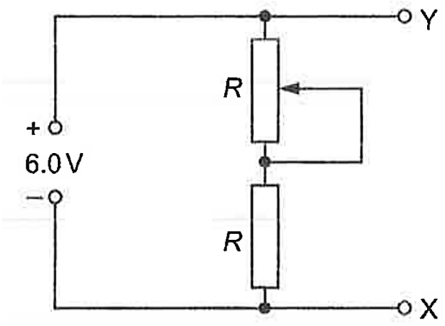
$V_X = 6.0 - k \times 3.0$

$V_{XY} = V_Y - V_X = 6.0 - (6.0 - 3.0k) = 3.0k$

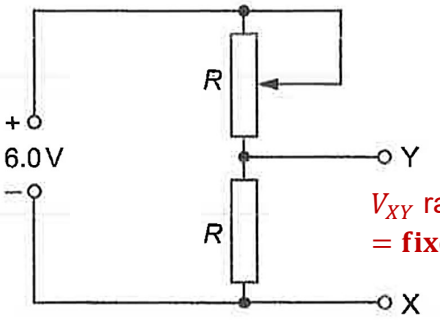
So, V_{XY} range = **0.0 V to 3.0 V**

C

V_{XY} range $= \mathbf{fixed\ at\ 6.0\ V}$

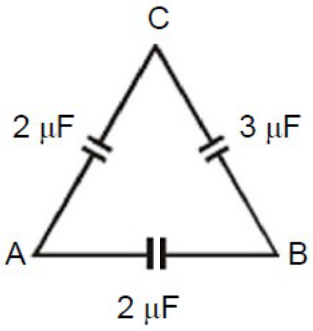


D



V_{XY} range $= \mathbf{fixed\ at\ 3.0\ V}$

21 Three capacitors are connected in the shape of a triangle ABC as shown in the figure below.



Ans: B

Since the 2 μF (AB) capacitor is in parallel with the series combination of 2 μF and 3 μF, the voltage across the entire AC-CB branch is also **5.0 V**.

For capacitors in series, the equivalent capacitance is:
$1/C_{eq} = 1/2 + 1/3 = 5/6 \rightarrow C_{eq} = 1.2\ \mu F$

Charge on each capacitor in the series branch:
$Q = C_{eq}V = 1.2 \times 5.0 = 6.0\ \mu C$

$V_{BC} = Q/C_{BC} = 6.0/3 = \mathbf{2.0\ V}$

A 5.0 V battery is connected between A and B.

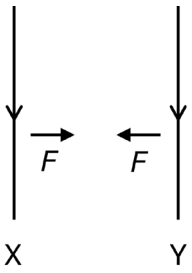
What is the voltage between B and C?

A 1.0 V **B** 2.0 V **C** 2.5 V **D** 3.0 V

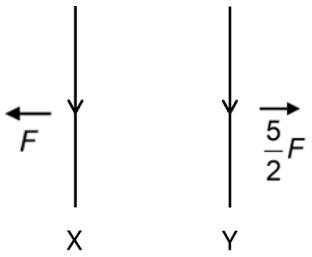
22 Two long, straight, current-carrying wires, X and Y, are parallel. The current in wire X is 2 A and the current in wire Y is 5 A.

Which diagram shows the correct combination of the magnitude of the forces and the directions of the current in wires X and Y?

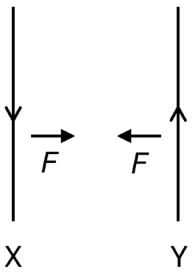
A



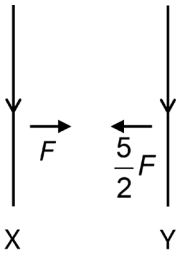
B



C



D



Force magnitude**:**

By Newton's Third Law, the forces on X and Y must be **equal in magnitude** — so both forces must be *F* (not 5/2 *F* on one and *F* on the other). This eliminates **B and D**.

Direction of forces:

- Currents in the **same direction** → wires **attract** (forces point toward each other)
- Currents in **opposite directions** → wires **repel** (forces point away from each other)

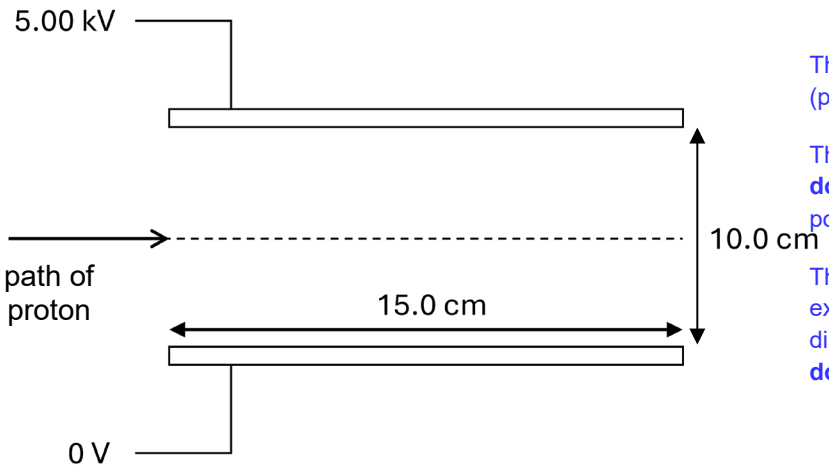
Checking A vs C:

Circuit **A**: Both currents flow downward (same direction) → wires should attract → forces point toward each other (X's force points right →, Y's force points left ←). This matches Circuit A. ✓

Circuit **C**: X flows downward, Y flows upward (opposite directions) → wires should repel → forces point away from each other. But Circuit C shows forces pointing toward each other — contradiction. ✗

Ans: A

23 A proton, initially travelling horizontally at 2.52×10^6 m s^{-1}, enters a uniform electric field at right angles to the field. The electric field is generated by applying a potential difference of 5.00 kV between two parallel conducting plates 15.0 cm long and 10.0 cm apart.



The top plate is at **5.00 kV** (positive) and the bottom at **0 V**.

The electric field points **downward** (from high to low potential).

The proton (positive charge) experiences a force in the direction of the field — **downwards**.

What is the change in height of the proton as it leaves the plates and the direction which it is deflected?

	change in height / m	direction of deflection
A	8.5×10^{-3}	downwards
B	8.5×10^{-3}	upwards
C	15.6×10^{-3}	downwards
D	15.6×10^{-3}	upwards

$$E = \frac{V}{d} = \frac{5000}{0.100} = 50000 \text{ V m}^{-1}$$

$$F = qE = 1.60 \times 10^{-19} \times 50000 = 8.00 \times 10^{-15} \text{ N}$$

$$a = \frac{F}{m} = \frac{8.00 \times 10^{-15}}{1.67 \times 10^{-27}} = 4.790 \times 10^{12} \text{ m s}^{-2}$$

$$t = \frac{L}{v_x} = \frac{0.150}{2.52 \times 10^6} = 5.952 \times 10^{-8} \text{ s}$$

$$y = \tfrac{1}{2}at^2 = \tfrac{1}{2} \times 4.790 \times 10^{12} \times (5.952 \times 10^{-8})^2 \Rightarrow y = 8.49 \times 10^{-3} \text{ m} \approx 8.5 \times 10^{-3} \text{ m}$$

Ans: A

24 Two point charges each of charge *Q* at a separation of *r* repel each other with a force *F*.

Each charge is increased by adding 3*Q* and the separation is increased to 2*r*.

What is the force between the charges?

A *F* B $9F/4$ C $4F$ D $8F$

Ans: C

Original force:

$$F = \frac{kQ \times Q}{r^2} = \frac{kQ^2}{r^2}$$

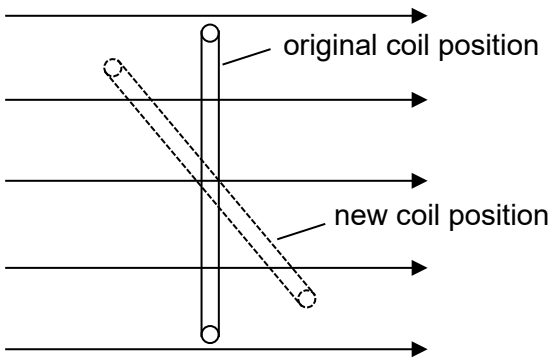
New force:

$$F_{new} = \frac{k \times 4Q \times 4Q}{(2r)^2} = \frac{16kQ^2}{4r^2} = \frac{4kQ^2}{r^2}$$

Ratio:

$$\frac{F_{new}}{F} = \frac{4kQ^2/r^2}{kQ^2/r^2} = 4 \Rightarrow F_{new} = 4F$$

25 A coil is situated with its plane perpendicular to a uniform magnetic field and within the field.



The coil is rotated to the new position shown.

How does the magnetic flux density and the total magnetic flux passing through the coil change during the rotation?

	magnetic flux density	total magnetic flux
A	decrease	decrease
B	increase	increase
C	no change	decrease
D	no change	increase

Ans: C

Magnetic flux density: Unchanged (external field is unaffected) → no change

Total magnetic flux: Decreases from maximum → decrease

26 A coil produces a magnetic flux of 0.85 Wb. A second coil of 1500 turns is wound on the same core, such that all the magnetic flux from the first coil links the second coil.

What is the e.m.f. induced across the terminals of the second coil when the flux produced by the first coil is reduced uniformly to zero in 0.19 s?

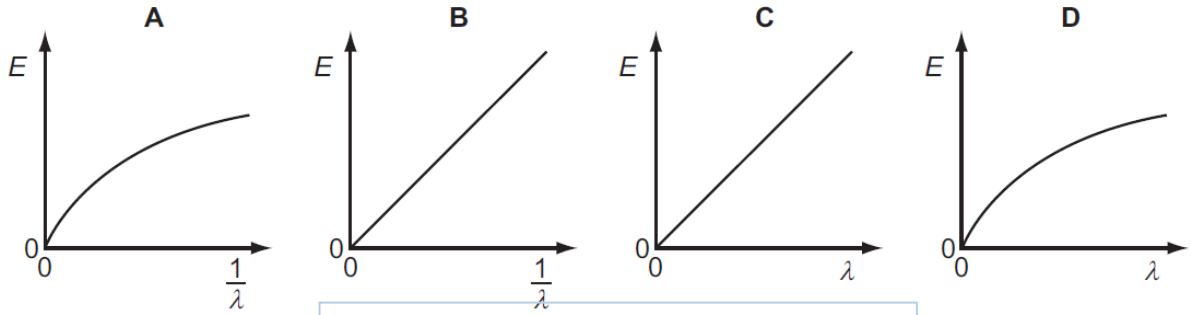
A 0 V **B** 4.47 V **C** 335 V **D** 6710 V

Ans: D

The induced e.m.f. in the second coil is given by: $\varepsilon = N\frac{\Delta\Phi}{\Delta t}$

where N is the number of turns, $\Delta\Phi$ is the change in flux, and Δt is the time taken. $\varepsilon = 1500 \times \frac{0.85}{0.19} = 6711\text{ V} \approx 6710\text{ V}$

27 Which graph shows the relationship between photon energy E and wavelength λ of electromagnetic waves?



Ans: B $E = hf = h\dfrac{c}{\lambda} \Rightarrow E \propto \dfrac{1}{\lambda}$

28 A particle of mass m in a well of width L has ground-state energy E.

A second particle has mass $2m$ and is confined in a well of width $2L$.

Its ground-state energy is:

A $E/2$

B $E/4$

C $E/8$

D $E/16$

Ans: C

Using: $E_n = \dfrac{h^2}{8mL^2}n^2$, we have

first particle ground state energy, $E = \dfrac{h^2}{8mL^2}(1)^2 = \dfrac{h^2}{8mL^2}$,

second particle ground state energy, $E' = \dfrac{h^2}{8(2m)(2L)^2}(1)^2 = \dfrac{1}{8}\left(\dfrac{h^2}{8mL^2}\right) = \dfrac{1}{8}E$

29 A sample of a radioactive isotope with nucleon number 210 decays by alpha emission into a stable isotope with nucleon number 206. The process has a half-life of 140 days.
If all the emitted alpha-particles escape from the sample but none of the daughter product escapes, what will be the percentage decrease in the mass of the source after 280 days?

A 0.48% **B** 1.43% **C** 25% **D** 75%

Ans: B

Assume the sample initially has number N unstable nuclei of nucleon number 210.

change in mass of the source after two half-lives:

$$= -\frac{3N}{4}(210) + \frac{3N}{4}(206) = -3N$$

fractional change in the mass of the source after two half-lives:

$$= \frac{-3N}{210N} = -\frac{1}{70} = -1.43\%$$

30 In medical radiotherapy, different types of radioactive emissions are chosen based on their penetrating abilities and ionising effects. Which combination correctly matches the emission type with its most appropriate medical application?

A alpha particles - external beam therapy for deep tumours

B beta particles - treating shallow skin cancers and surface tumours

C gamma rays - treating only surface-level conditions

D alpha particles - medical imaging of internal organs

Ans: B

A: This is incorrect because alpha particles have extremely poor penetrating ability and are completely stopped by the skin's outer layer. They cannot reach deep tumours from an external source, making external beam therapy impossible. All the radiation would be absorbed at the skin surface without reaching the intended target.
B: This is correct because beta particles possess moderate penetrating ability, allowing them to penetrate several millimetres into tissue whilst delivering effective ionising radiation to surface and near-surface tumours. Their limited range prevents excessive damage to deeper healthy tissues, making them ideal for treating skin cancers and shallow tumours.
C: This is incorrect because gamma rays have very high penetrating ability and would largely pass through surface tissues without depositing sufficient energy for treatment. Using such penetrating radiation for surface conditions would expose deeper healthy tissues to unnecessary radiation whilst providing poor treatment efficacy at the target site.
D: This is incorrect because alpha particles cannot penetrate body tissues from external sources and cannot exit the body to reach detectors when administered internally. Medical imaging requires radiation that can either penetrate the body or travel from internal sources to external detectors, which alpha particles cannot accomplish due to their extremely limited range.

BLANK PAGE